Codex Build Brief: All MUSTs, Minimal Setup

Oct 25, 2026 · @Varizz

1. What changed and the setup promise

This brief replaces the architecture and scope of the earlier "Minimal Demo Plan". Its guardrails carry over and are repeated here so this document stands alone. You will give it to Codex, so the **scope is now every Must story in the SRS**, including ones I had suggested dropping (`US-H05`, `US-H07`, `US-B04` and others). The real constraint is **setup on other machines**.

The setup promise. A new machine needs only:

1. **PostgreSQL** (any recent version, plain install, no extensions)
2. **Ollama** (with the two models, pulled once)
3. **Python 3.11 or newer** with a virtual environment
4. **The frontend**, shipped **already built** inside the repo, so Node is not needed on demo machines. Node is needed only on the machine that builds it (yours, or Codex's)

Nothing else: no Docker, Redis, Celery, message queue, pgvector, Tesseract, SMTP server, cloud account or internet access during the demo itself.

Earlier plan item	Now
Celery, Redis, Celery Beat	A `jobs` table, a small in-process worker thread pool, and APScheduler (pure Python) for the few timed tasks
pgvector extension	**Not used.** Embeddings stored as a plain array column and compared in Python (numpy), always filtered by workspace first. Fine at demo scale, and it removes the one Postgres extension that causes install trouble
Docker Compose	**Not used.** One command starts everything
Alembic	A simple versioned migration runner inside the app (or `create_all` plus a `schema_version` table)

Earlier plan item	Now
Live email	A **mail backend interface**: default is an in-app outbox page showing every email and its link; real SMTP is optional through `.env`
Node on every machine	Frontend built once and served by the backend as static files
Scanned-PDF OCR via system Tesseract	Optional pip-only extra, off by default (section 4)
PDF export needing system libraries	`reportlab` with a bundled Japanese font, pip only (section 4)

Why this is safe for the SRS: none of these choices changes a requirement. The SRS asks for outcomes (isolation, provenance, confirmations), and section 8 says providers should be swappable. It does not ask for Celery, Redis or pgvector.

2. Zero-headache setup design

2.1 The commands a person on a new machine runs

1. Install PostgreSQL, Ollama and Python 3.11+ (one-time).
2. In the repo folder: `python -m venv .venv`, activate it, then `pip install -r requirements.txt`.
3. Copy `.env.example` to `.env` and set the database URL (and, only if needed, the model names).
4. `python manage.py setup`
5. `python manage.py run` (starts the app and opens the browser)

Two more commands help with a distributed demo:

- `python manage.py check`: a preflight report (section 2.3).
- `python manage.py seed --reset`: restores the demo data to its starting state before each presentation.

Use **one Python entry point** (`manage.py`) rather than shell scripts, so the same commands work on Windows, macOS and Linux.

2.2 What `setup` must do (idempotent, safe to run twice)

- Check that PostgreSQL is reachable with the URL in `.env`; create the database if it is missing; if the role lacks permission, print the exact one-line fix.
- Create or migrate all tables.
- Ask Ollama to pull the generation and embedding models if they are not already installed, with visible progress. Never fail silently.
- Load the seed data: two companies, four logins, library cards, templates, workflow B1, fixtures (section 2.4).
- Print the login details and the URL.

2.3 What `check` must report

Check	Pass means
PostgreSQL	Connects; schema version current
Ollama	Reachable; both models installed
Generation smoke test	One short Japanese and one English answer return; shows time to first token and tokens per second
Embedding smoke test	A test sentence embeds; vector size recorded
Frontend	Built files found
Seed data	Demo companies and logins exist
Disk	Upload and export folders writable

The same report should be visible at `/health` for a presenter who wants to confirm the machine is ready. Because machines differ, **speed varies by machine**, so show the measured numbers rather than assuming them.

2.4 Rules for Codex (setup and portability)

- **One process.** The API, the static frontend, the job workers and the timed tasks run together. Do not require a second terminal.
- **All settings in `.env`:** database URL, Ollama base URL, generation model, embedding model, ports, mail backend, upload folder, budgets. Nothing hard-coded.

- **Relative paths only.** One `data/` folder inside the repo for uploads, exports and caches, with a subfolder per workspace.
- **No outside network needed at demo time.** No CDN fonts or scripts. Bundle fonts and all frontend assets. Ship **offline fixtures**: a saved copy of the demo website's pages so the crawl step can run from disk when the machine is offline, as well as live when it is not.
- **Japanese fonts.** Bundle a Japanese web font (an open-licence font such as Noto Sans JP) so the interface looks the same on every machine, and use the same font for PDF export.
- **Deterministic, resettable seed data.** Fixed IDs for demo companies so scripts and screenshots do not change.
- **Clear first-run errors.** If Postgres or Ollama is missing, the app shows a plain message with the next step, never a stack trace.
- **Ports and host:** default to `localhost` only. The Ollama server is never exposed to the browser or to crawled content.
- **Windows friendliness:** no shell-specific syntax in code or docs; test path handling with spaces.
- **Model files for offline machines:** Ollama's model folder can usually be copied between machines, which avoids a multi-gigabyte download at the venue. Do not rely on venue Wi-Fi for the first model download.

3. Scope: every Must story

Counts, from the SRS priority column: there are **67 Must stories**. SRS section 11 also puts **six Should stories** in v1 because it includes whole epics (A, C and E): `US-A06`, `A08`, `C07`, `C09`, `E06`, `E08`. That makes **73 stories** to build. The exact acceptance criteria for each are in SRS section 4 and **the SRS governs**. The tables below give a one-line meaning and the lean implementation note that avoids extra infrastructure.

Codex's own plan marks five Must stories as fully deferred (`US-B04`, `US-H05`, `US-H07`, `US-I06`, `US-J01`), four of the six in-v1 Should stories as deferred (`US-A06`, `A08`, `C07`, `C09`), and many other Musts as only partly done (for example `US-F01` OCR and images, `US-E05` images and PowerPoint, `US-F06` PDF). All of these are now in scope, so the coverage table in Codex's plan must be redone.

3.1 Epic A: access, tenancy, roles (SRS 4.1)

ID	Meaning	Lean implementation note
`US-A01` Must	Super Admin creates a workspace	Unique slug; status Invited; audit entry
`US-A02` Must	Invite Company Admin by email	7-day single-use token; resend and revoke; **mail backend** with in-app outbox as default
`US-A03` Must	Accept invite, set password (12+ characters), choose language	Wizard opens on first entry
`US-A04` Must	Company Admin invites Members	Same invitation flow
`US-A05` Must	Sign-in, sign-out, reset (1 hour), 12-hour idle timeout, rate limit	Server-side sessions in Postgres; failed attempts counted in the database
`US-A06` Should (in v1)	MFA with authenticator app, forced for admin roles	`pyotp`; recovery codes. Add `.env` flag `DEMO_SKIP_MFA` for demo logins, default on, with the real flow still built and testable
`US-A07` Must	Workspace isolation	One data-access layer; see guardrail G1
`US-A08` Should (in v1)	Company Admin grants a Coordinator read or comment access	Grant and revoke in settings; audit entry
`US-A09` Must	Switch Japanese / English	Interface, emails, AI replies; one reply in the other language on request
`US-A10` Must	Deactivate and reactivate a workspace	Block sign-in; keep data

3.2 Epic B: admin console (SRS 4.2)

ID	Meaning	Lean implementation note
`US-B01` Must	Companies list with statuses, filter, sort, CSV	Dormant computed from last activity at read time (30 days)
`US-B02` Must	Company record with shared sections only	Counts, never content
`US-B03` Must	Portfolio dashboard	Aggregates from SQL queries; charts drawn with a bundled local chart library
`US-B04` Must	Use-case library authoring	Create and edit with Japanese and English fields, stage, entry point, horizon, autonomy level, data needed; states draft, review, published; version history; **publishing blocked until both languages exist** (`US-K05`); Super Admin can review and comment
`US-B06` Must	Feedback inbox	Statuses New, In review, Addressed; tags; reply; export (CSV)
`US-B07` Must	Audit log	Insert-only; filters by actor, company, date; 1-year retention value stored (clean-up job optional)

3.3 Epic C: onboarding and company profile (SRS 4.3)

ID	Meaning	Lean implementation note
`US-C01` Must	Onboarding wizard	Resumable; website, product text, catalogue and photo uploads, basic facts
`US-C02` Must	Website analysis	`httpx` plus an HTML parser; `robots.txt` via the standard library; depth and page limits; block private addresses and re-check redirects; progress via the job record; offline fixtures

ID	Meaning	Lean implementation note
`US-C03` Must	Product analysis	Structured output validated against a schema; add, merge, rename, delete products
`US-C04` Must	Eight-section profile with Confirm / Edit / Reject and history	Fact ledger (section 5)
`US-C05` Must	Completeness and gaps	A written list of required fields per section; gap links to a workflow or question
`US-C06` Must	Facts from chats, answers and documents flow in after review	Single `propose_fact` function
`US-C07` Should (in v1)	Re-run website analysis with a diff	Never overwrite confirmed facts without review
`US-C08` Must	Choose which sections the Operator sees	Default overview and products summary
`US-C09` Should (in v1)	Export profile as DOCX and PDF, in Japanese or English	As-of date and provenance legend
`US-C10` Must	Source content is data, not instructions	Guardrail G9

3.4 Epic D: workflows (SRS 4.4)

ID	Meaning	Lean implementation note
`US-D01` Must	AI proposes the first workflows	Company discovery, Product deep-dive per main product, Market and product-market fit, Manufacturing and AI readiness; start, defer, dismiss
`US-D02` Must	Steps with type, status, owner, due date	Build **one generic workflow engine**; define templates B1 to B4 as data. B5 to B8 are extra data entries if time allows. Run B1 and B3 end to end

ID	Meaning	Lean implementation note
`US-D03` Must	Adaptive question sets of 5 to 10	"I don't know" and "assign to a colleague"; answers become Stated facts in the queue; text, upload (voice excluded)
`US-D04` Must	Stepper and dependency view	Simple graph of what each step feeds
`US-D05` Must	Each workflow has its own chat; step changes need confirmation	Pending action (guardrail G2)
`US-D08` Must	Completed workflow delivers outputs and proposes next ones	Output types defined in the template data

3.5 Epic E: chats (SRS 4.5)

ID	Meaning	Lean implementation note
`US-E01` Must	Chat list: search, pin, rename, archive; start blank, from template, workflow or document	Search with `ILIKE` over titles and message text
`US-E02` Must	Streaming reply with citations and message actions	Server-sent events. Actions include **turn into workflow**: this drafts a workflow from the conversation, behind a confirmation. It is the core of `US-D09`, which is otherwise not in v1
`US-E03` Must	Context panel with toggles and without-profile notice	Toggles change the real prompt (test 3, section 5)
`US-E05` Must	Attachments up to 25 MB: PDF, images, spreadsheets, Word, PowerPoint	`pypdf`, `python-docx`, `openpyxl`, `python-pptx`; images in section 4
`US-E06` Should (in v1)	Suggested prompts	From open gaps, waiting steps and relevant library cards; dismissible
`US-E07` Must	Facts stated in chat go to the review queue	Guardrail G7

ID	Meaning	Lean implementation note
`US-E08` Should (in v1)	Share a chat and export as Markdown or PDF	PDF via section 4
`US-E09` Must	Monthly token budget, notice at 80%	Counter from the `model_calls` table by calendar month; no reset job needed
`US-E10` Must	Long chats summarised, summary visible	Summarise older turns when the history grows past a limit

3.6 Epic F: knowledge base and document studio (SRS 4.6)

ID	Meaning	Lean implementation note
`US-F01` Must	Add files and web pages as sources	PDF, DOCX, XLSX, PPTX, images, text, URLs; states Processing, Ready, Failed; auto tags; 1 GB per workspace, configurable; **OCR and images: section 4**
`US-F02` Must	Source view with extracted text, summary, detected facts, re-index, delete	Deleted source is excluded from retrieval immediately and its dependent facts return to review
`US-F03` Must	Generate a document from a prompt and template	Choose context (profile, sources, workflow answers), language, length; request saved
`US-F05` Must	Editor with AI actions	Rich text editor bundled locally (for example TipTap); rewrite, shorten, expand, translate JA/EN, insert profile facts, table from sources, comments, versions with compare and restore
`US-F06` Must	Export DOCX, PDF, Markdown	`python-docx`, `reportlab` with bundled font, plain Markdown
`US-F07` Must	Save a finished document as a source	Indexes it and shows its origin

ID	Meaning	Lean implementation note
`US-F09` Must	Every statement shows sourced or "AI assumption, verify"	Export can include or exclude markers
`US-F04` Should (not in v1 list)	Outline approval	Cheap, include if time allows

3.7 Epic H: AI use-case explorer (SRS 4.8)

ID	Meaning	Lean implementation note
`US-H01` Must	Method explainer under 15 minutes	Content in both languages; see content gap below
`US-H02` Must	Clickable factory map	Seven stages plus lanes; stages from the profile highlighted; typical operations, systems and examples per stage; see content gap below
`US-H03` Must	Library cards with search and filters	Seed all 44 Appendix A rows with bilingual title, stage, entry point, horizon, level and data needed. Write the remaining card fields (problem solved, how it works, systems touched, prerequisites, what good looks like) for at least the ten demo cards, marked as drafts. Filters by stage, entry point, horizon, level. Value shown only as observed ranges with no client named
`US-H04` Must	Fact-linked recommendations	Ranked list with a Why tied to profile facts and readiness gaps; one-click add; feedback on each

ID	Meaning	Lean implementation note
`US-H05` Must	**Custom use-case wizard and charter**	Fields: problem, stage, who is affected, how data is captured today (paper, spreadsheet, system, sensor), systems involved, expected outcome and KPI. AI drafts a one-page charter, classifies the step and the autonomy level, lists prerequisites and data needs. User edits and saves. A "Make a charter document" button uses the AI use-case charter template
`US-H07` Must	**Road map board**	2x2 of business impact against readiness to deploy with quadrants Quick wins, Strategic bets, Momentum builders, Later; a programme view by horizon H1 to H3 on a multi-year line; drag and drop; dependencies shown as "needs the data created by"; export as a document (Factory AI road map template). Because the readiness check (`US-H06`) is v1.1, give each use case a **simple readiness rating on the four dimensions (data, process, people, systems)**, suggested by the AI, labelled Inferred, and editable
`US-H10` Must	Learning content per stage	Short explainer, anonymised example and "what we need from the plant" list, plus a glossary, in both languages

Content gap Codex cannot fill from the SRS. `US-H01` , `H02` and `H10` depend on Atomic Loops material that the SRS only cites (the Assessment deck and the Factory Assessment Framework). The SRS names the five audit stages, four steps, L0 to L5 names, three horizons and four value principles, but **does not list the "three outcomes of a report"** and gives typical operations and systems only for the Assembly example. Have Codex draft the rest and mark it "draft pending Atomic Loops review". If you have the two Atomic Loops documents, give them to Codex.

3.8 Epic I: feedback (SRS 4.9)

ID	Meaning	Lean implementation note
`US-I01` Must	Thumbs, reason and text on every AI output	Messages, facts, documents, recommendations, charters
`US-I02` Must	Feedback widget on every page	Categories bug, idea, confusing, wrong, praise; **screenshot upload** (a file input is enough); page, role, language captured
`US-I03` Must	Report a wrong fact	Fact becomes Disputed; correct value requested; to review queue and inbox
`US-I04` Must	Users see what happened to their feedback	Statuses Received, In review, Addressed; replies
`US-I05` Must	Feedback analytics	Volume, positive rate, by feature and by company in aggregate, trend, export
`US-I06` Must	Evaluation set export	Only where the company's consent setting allows; never for third-party training; personal data removed first (basic redaction of emails, phone numbers and names)

3.9 Epic J: notifications, settings, help (SRS 4.10)

ID	Meaning	Lean implementation note
`US-J01` Must	Notifications and email digest	In-app notification centre; events: invitation, step waiting on me, document ready, feedback answered (announcement event not needed because `US-B09` is out); per-user preferences; digest generated by a timed task inside the app (APScheduler) and delivered through the mail backend

ID	Meaning	Lean implementation note
`US-J03` Must	Workspace settings	Name, logo, default language, members, Coordinator access, sharing with the Operator, retention, **export of all data as a ZIP** (documents, profile, chats as JSON and Markdown), deletion request
`US-J04` Must	Guided tour and help	First-run tour, bilingual help articles, contextual tips, onboarding checklist that ticks itself off
`US-J06` Must	Consent records	Terms and privacy acceptance at first sign-in; separate settings for Operator visibility and feedback use, each with history. Legal text is placeholder, marked draft (`OQ-14`)

3.10 Epic K: Japanese first, English equal (SRS 4.11)

ID	Meaning	Lean implementation note
`US-K01` Must	No untranslated screen	Resource files per language; a test that fails when a key is missing; toggle on every page
`US-K02` Must	AI replies in the user's language	Handle mixed-language messages
`US-K03` Must	Generated output in workspace language, other language on demand	One-click version in the other language; glossary kept consistent (Appendix F)
`US-K04` Must	Read Japanese sites and documents correctly	Encodings; store facts in the source language with a "translated" label
`US-K05` Must	Bilingual library by construction	Publish blocked until both languages exist; Operator reviewer approves the Japanese (`OQ-10`)
`US-K06` Must	Locale-correct dashboards and charts	Numbers, dates, yen; ISO dates in admin exports

ID	Meaning	Lean implementation note
`US-K09` Must	Japanese typing and reading	IME composition in every field; line-breaking; fonts
`US-K11` Must	Emails and notifications in the user's language	Templates in both languages
`US-K07` Should	Cross-language search	Comes almost free with a multilingual embedding model; include
`US-K10` Should	Operator reads shared content in own language	Optional, marked "translated by AI"

3.11 Not built (and say so)

Voice (`US-E04` , `K08`), planning canvases (Epic G), Admin Copilot (`US-B08`), announcements (`US-B09`), programme matching (`US-B10`), template and question-bank editor (`US-B05`), custom workflow builder (`US-D06`), reminders and pause (`US-D07`), template authoring (`US-F08`), whole-document translation (`US-F10`), readiness check as a separate assessment (`US-H06`), lite assessment and assessment request (`US-H08` , `H09`), quarterly survey (`US-I07`), activity feed (`US-J02`), workspace-wide search (`US-J05`), SSO and public API. These are Could or Won't in the SRS, or Should stories that SRS section 11 places in v1.1 or v2.